

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Supplementary Examination – 2023

Course: B. Tech. Branch: Electronics & Telecomm. Engg. Semester: V
Subject Code & Name: BTETOE505C/BTEXOE505C Optimization Techniques
Max Marks: 60 Date: 19.08.2023 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Provide examples of real-world engineering problems that can be solved using optimization techniques.	(L1/CO1)	6
B) Discuss the significance of formulating a clear and appropriate objective function in an optimization problem. Provide examples to illustrate the importance of well-defined objectives.	(L1/CO2)	6
C) Define constraints in the context of optimization problems. How do constraints impact the feasible solution space?	(L1/CO1)	6
Q.2 Solve Any Two of the following.		12
A) Explain the basic principles of gradient-based optimization methods, such as gradient descent. How do these methods perform in non-convex optimization landscapes?	(L2/CO2)	6
B) Explain the concept of convexity and concavity for functions of one variable. What are the key characteristics of convex and concave functions?	(L1/CO1)	6
C) Explain the challenges associated with optimizing functions of multiple variables.	(L1/CO1)	6
Q. 3 Solve Any Two of the following.		12
A) Define the concept of equality constraints in optimization problems. How can these constraints affect the feasible solution space?	(L2/CO2)	6
B) Define the Hessian matrix and its significance in optimization. How does the Hessian matrix relate to the curvature of the function's surface?	(L2/CO1)	6
C) Define the Kuhn-Tucker conditions and their role in solving constrained optimization problems. How do they extend the Lagrangian method to handle inequality constraints?	(L1/CO1)	6

Q.4 Solve Any Two of the following.		12
A) Define the standard form of a linear programming (LP) problem. What are the key characteristics that distinguish a problem in standard form?	(L1/CO2)	6
B) Discuss the advantages and disadvantages of using the canonical form for solving LP problems.	(L1/CO1)	6
C) Explain the graphical method for solving a two-variable linear programming problem.	(L2/CO1)	6
Q. 5 Solve Any Two of the following.		12
A) Define sequential optimization. How does it differ from single-stage optimization, and what types of problems does it address?	(L1/CO2)	6
B) Differentiate between deterministic and stochastic multistage decision problems. How does uncertainty influence the decision-making process in each case?	(L2/CO1)	6
C) Define forward and backward recursions in the context of dynamic programming. How do these recursive equations help in solving multistage decision problems?	(L1/CO1)	6

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